Code No: 126EQ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
OBJECT ORIENTED ANALYSIS AND DESIGN
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the phases in the unified process?	[2]
b)	List the UML diagrams.	[3]
c)	Illustrate the visibility feature for a class.	[2]
d)	What is the content of a class diagram?	[3]
e)	Illustrate use case organization.	[2]
f)	What are the standard stereotypes associated with links?	[3]
g)	How to represent thread synchronization?	[2]
h)	Illustrate shallow history state.	[3]
i)	List management artifacts.	[2]
j)	What is meant by framework?	[3]

PART – B

(50 Marks)

2. Describe the importance of modeling and principles of object-oriented modeling. [10]

OR

3.a) What is a workflow? Discuss workflows in software development life cycle.
b) Illustrate grouping and annotational things of UML. [5+5]

4.a) How to model the semantics of a class? Explain with an example.
b) Differentiate between aggregation and composition. [5+5]

OR

5. Construct a class diagram and an object diagram for hostel management system. [10]

6.a) What is the need of interactions? What is the role of messages in interactions?
b) Explain the significance of collaboration diagram in modeling. [5+5]

OR

7.a) What is the difference between activity state and action state.
b) How to model requirements of a system using use case diagram? [5+5]

8.a) Discuss the common uses of state machines and state chart diagrams.
b) Demonstrate modeling family of signals. [5+5]

OR

9.a) With suitable examples, describe different types of components.
b) Draw a node diagram to model a fully distributed system. [5+5]

10. Describe any five types of design patterns used in object-oriented analysis and design. [10]

OR

11. Perform modeling process view and use case view of unified library application. [10]

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